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Investigations of swirl flames in a gas turbine model combustor II. Turbulence–chemistry interactions

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Abstract

The thermochemical states of three swirling CH_4/air diffusion flames, stabilized in a gas turbine model combustor, were investigated using laser Raman scattering. The flames were operated at different thermal powers and air/fuel ratios and exhibited different flame behavior with respect to flame instabilities. They had previously been characterized with respect to their flame structures, velocity fields, and mean values of temperature, major species concentrations, and mixture fraction. The single-pulse multispecies measurements presented in this article revealed very rapid mixing of fuel and air, accompanied by strong effects of turbulence–chemistry interactions in the form of local flame extinction and ignition delay. Flame stabilization is accomplished mainly by hot and relatively fuel-rich combustion products, which are transported back to the flame root within an inner recirculation zone. The flames are not attached to the fuel nozzle, and are stabilized approximately 10 mm above the fuel nozzle, where fuel and air are partially premixed before ignition. The mixing and reaction progress in this area are discussed in detail. The flames are short (<50 mm), especially that exhibiting thermoacoustic oscillations, and reach a thermochemical state close to adiabatic equilibrium at the flame tip. The main goals of this article are to outline results that yield deeper insight into the combustion of gas turbine flames and to establish an experimental database for the validation of numerical models.
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Keywords: Gas turbine model combustor; Swirl; Turbulence–chemistry interaction; Finite-rate chemistry; Lean blowoff; Validation measurements

1. Introduction

A major goal in the development of new gas turbine (GT) combustion concepts is the reduction of pollutant emissions, especially NO_x. This can be achieved by very fuel-lean combustion, either in premixed flames or in fast-mixing diffusion flames where regions with near-stoichiometric reacting mixtures

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are avoided or kept small. However, these flames are prone to instabilities in the form of unstable ignition or thermoacoustic oscillations. To gain deeper insight into the physical and chemical processes governing flame behavior, the German Aerospace Center (DLR) has established a laboratory-scale GT combustor with good optical access which enables the application of various laser diagnostic techniques. Three overall lean CH_4/air diffusion flames with different flame behavior have been investigated in this burner. The flow field was measured by laser Doppler velocimetry (LDV), the flame structures by planar laser-induced fluorescence (PLIF), and the joint probability density functions (PDFs) of temperature, major species concentrations, and mixture fraction by laser Raman scattering. In the preceding article [1], the general flame characteristics were described; these include structures, flow field, heat release, and mean and rms values of mixture fraction, temperature, and species concentrations. The main subjects of this article are the effects of turbulence–chemistry interactions in the flames which are revealed by the correlations between different quantities measured by laser Raman scattering. One main goal of the work was a detailed characterization of GT-like flames to gain a better understanding of processes like mixing and flame stabilization, finite-rate chemistry effects, and effects of unsteady combustion behavior. A second goal was the establishment of an experimental database that can be used for the validation of numerical flame calculations.

In the flames investigated, finite-rate chemistry effects play an important role and manifest themselves, for instance, in the form of local flame extinction or ignition delay. These effects have, of course, a strong influence on the tuning range of stable operation of the combustor, especially on the lean blowoff limit and the noise level. Furthermore, finite-rate chemistry effects can change the emissions of pollutants, e.g., NO_x, CO, and unburned hydrocarbons. With respect to numerical flame simulations, turbulence–chemistry interactions represent a difficult combustion regime which requires sophisticated chemical submodels. Here, a validation of results from model calculations by experiments is of special importance.

A detailed experimental analysis of finite-rate chemistry effects in turbulent flames requires the simultaneous detection of as many quantities as possible, and if possible, with multidimensional spatial resolution. Laser Raman scattering enables the simultaneous detection of the concentrations of all major species (O_2, N_2, CH_4, H_2, CO, CO_2, H_2O), mixture fraction, and temperature and is therefore the method of choice for characterization of the thermochemical state of a flame [2–4]. In some cases, Raman and LIF systems have been combined so that additional species like NO and OH could be detected [5–11]. By use of these techniques, a number of turbulent jet flames (free, piloted, bluff-body, or swirl-stabilized) have been investigated [12–18] and many of them are well-documented in the data archives of the TNF Workshop [19]. Those studies exemplified the difficulties of numerical flame models with the treatment of intense local flame extinction and reignition. Even stronger effects of turbulence–chemistry interaction have been documented in a swirl burner at the University of Sydney [20] and in the TECFLAM swirl burner [21–23]. In contrast to jet flames, flames with recirculation zones exhibit regions of intense mixing of hot combustion products with unburned gas. Depending on the temperature and gas composition, the time for ignition of such mixtures (ignition delay time τ) is in the range of <1 ms to many milliseconds and can therefore be on the same order as the time scale for convective transport and turbulent mixing. In addition to localized flame extinction, this presents another dominant effect of turbulence–chemistry interaction in the flames under investigation.

In the current article, the thermochemical states of three swirling CH_4/air diffusion flames stabilized on a GT nozzle are described and compared: flame A, with a thermal power P of 35 kW and an overall equivalence ratio Φ_{glob} of 0.65, burned stably; flame B, with $P = 10$ kW and $\Phi_{glob} = 0.75$, exhibited self-excited thermoacoustic oscillations with a frequency f of $\approx$290 Hz; and flame C operated close to the lean extinction limit with $P = 7.6$ kW and $\Phi_{glob} = 0.55$. Effects of mixing, flame stabilization, finite-rate chemistry, and reaction progress are addressed in the article. The thermoacoustic oscillations of flame B have also been analyzed by phase-resolved measurements [24,25], and some of the findings from those investigations are used here to support the discussion of the characteristics of flame B.

2. Experimental

2.1. Combustor and flames

The combustor and the flames were described in the preceding article [1] and only a short resume is given here. Fig. 1 is a schematic drawing of the GT model combustor. Coswirling dry air at room temperature is supplied to the flame through a central nozzle (diameter 15 mm) and an annular nozzle (i.d. 17 mm, o.d. 25 mm contoured to an o.d. of 40 mm), and nonswirling CH_4 is fed through 72 channels (0.5×0.5 mm), forming a ring between the air nozzles. The exit planes of the fuel and central air nozzles are located 4.5 mm below the exit plane of the outer

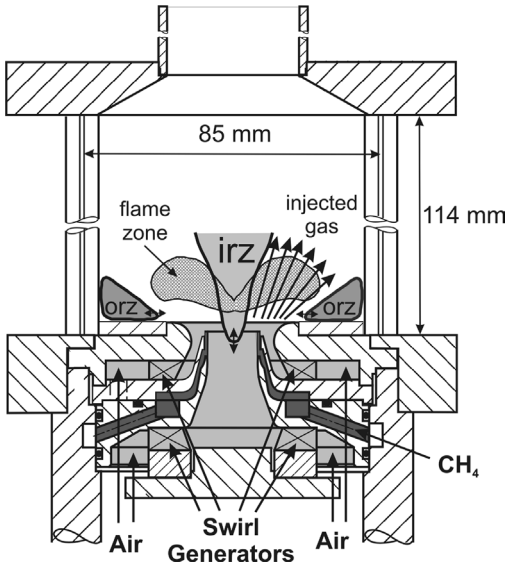

Fig. 1. Schematic drawing of the burner and combustion chamber. The flame zone location and the main features of the flow field, i.e., stream of injected gases, inner recirculation zone (irz), and outer recirculation zone (orz), are also indicated.

air nozzle, which is defined as height $h = 0$. The combustion chamber has a square section of 85 × 85 mm and a height of 114 mm and is equipped with four quartz plates held by steel posts (diam 10 mm) in the corners. A conical top plate made of steel with a central exhaust tube (diam 40 mm, length 50 mm) forms the exit.

The operating conditions of the three flames investigated are summarized in Table 1. Here, the index *global* refers to the mixtures resulting from the mass flow rates. The mass flows of the gases were controlled by Brooks flow controllers (5853S for air and 5851S for CH_4) with an accuracy of typically ±0.5%. The flames appeared as type 2 swirl flames [26] with a cone-shaped toroidal flame zone with different opening angles and with pronounced recirculation zones on the axis (inner recirculation zone, irz) and near the walls of the combustion chamber (outer recirculation zone, orz), as indicated in Fig. 1. The swirl numbers taken from the lowest velocity profiles at $h \approx 1$ mm are about $S = 0.9$ for flame A, and about $S = 0.55$ for flames B and C. The nozzle Reynolds numbers, derived from the cold inflow, are 15,000 for flames B and C and 58,000 for flame A. Flame A burned stably and emitted broadband noise with a peak at $f \approx 380$ Hz. Flame B exhibited the highest sound level with a quite discrete frequency of about 290 Hz. Flame C, operated near the lean extinction limit, burned unstably with random fluctuations of the stabilization zone in the form of varying liftoff height.

2.2. Joint PDF measurements by laser Raman scattering

Simultaneous measurement of the concentrations of the major species (O_2, N_2, CH_4, H_2, CO, CO_2, H_2O), the mixture fraction, and the temperature was accomplished by laser Raman scattering [22]. The radiation of a flashlamp-pumped dye laser (Candela LFDL 20, wavelength $\lambda = 489$ nm, pulse energy $E_p \approx 3$ J, pulse duration $\tau_p \approx 3$ μs) was focused into the combustion chamber, and the Raman scattering emitted from the measuring volume (length 0.6 mm, diam 0.6 mm) was collected by an achromatic lens ($D = 80$ mm, $f = 160$ mm) and relayed to the entrance slit of a spectrograph. The dispersed and spatially separated signals from the different species were detected by photomultiplier tubes (PMTs) in the exit plane of the spectrograph and electronically sampled by gated boxcar integrators. The species number densities were calculated from these signals using calibration measurements [27,28], and the temperature was deduced from the total number density via the ideal gas law. At each measuring location within a scanning pattern of roughly 100 points, 500 single-pulse measurements were performed from which the joint probability density functions (PDFs) were established.

Signal interferences from C_2 and PAHs were very low in the flames investigated and there was hardly any need to apply correction procedures as, for instance, in flames with long residence times under fuel-rich conditions [17,29–33]. Nonetheless, the signal background was recorded by additional PMTs in Raman-free spectral regions, and a few samples with

Table 1
Parameters of the three flames investigated

	Air		CH_4		P_{th}[a] (kW)	Φ_{glob}	f_{glob}	$T_{glob\ ad}$ (K)
	sl/min	g/min	sl/min	g/min				
A	850	1095	58.2	41.8	34.9	0.65	0.0365	1750
B	218	281	17.2	12.3	10.3	0.75	0.0418	1915
C	218	281	12.6	9.0	7.6	0.55	0.0310	1570

[a] P_{th}, thermal power; Φ_{glob}, equivalence ratio for the overall mixture; f_{glob}, mixture fraction for the overall mixture; $T_{glob\ ad}$, adiabatic temperature for the overall mixture with $T_0 = 295$ K.

significant background were filtered out in the data reduction routine. Also, single shots with PMT saturation or unrealistic C/H or N/O ratios were also discarded during data evaluation. In the worst case, the percentage of discarded samples was 6.2% (at $h = 10$ mm, $r = 9$ mm) in flame B and below 5% for flames A and C. On average, the fraction of filtered samples was on the order of 1%. A significant bias of the joint PDFs due to filtering was not noted.

With respect to measurement uncertainties, one has to distinguish between systematic errors arising from, for example, uncertainties of the calibration or flowmeters, and statistical errors, which are caused mainly by the statistics (shot noise) of the detected Raman photons N_P in a single-shot measurement. Systematic uncertainties were typically 3% for temperature and 3–4% of the mole fractions of O_2, H_2O, CO_2, and CH_4. The systematic errors are largely independent of the mole fraction and temperature in the flame. Statistical uncertainties were quantified by recording single-shot data sets in stable laminar flames. The rms fluctuations in these flames were, for example, 2.5% for T at $T = 1916$ K, 3.2% for H_2O at a mole fraction of $X(H_2O) = 0.19$, 7% for O_2 at $X(O_2) = 0.06$, 7.4% for CO_2 at $X(CO_2) = 0.068$, and 1% for the mixture fraction. In an electrically heated flow of pure CH_4 the rms value was 1% at 850 K. To a good approximation, the rms fluctuations scale with $N_P^{-0.5}$, so that the rms value increases by a factor of 2 when the mole fraction decreases by a factor of 4 (at constant temperature).

2.3. Spatial averaging effects

If fluid elements with different temperatures and gas compositions, e.g., fuel-lean exhaust gas and unburned fuel, are present in different parts of the measuring volume (i.e., not mixed on a molecular level), the Raman measurement yields an average temperature and gas composition. Such a result can, in principle, mimic a nonequilibrium thermochemical state although both gas parcels are at equilibrium. To estimate whether spatial averaging effects occurred in the presented Raman results, spatial gradients within the flame are considered. Two-dimensional temperature or major species concentration measurements are difficult and have not been performed in this study. However, the 2-D OH LIF measurements reported in [1] can be exploited for this purpose. They were performed with a spatial resolution better than 0.3 mm. Due to the exponential dependence of OH concentration on temperature, spatial gradients in the OH distributions are generally higher than temperature gradients and the steepest gradients are expected in reaction zones. Analysis of the OH PLIF images from flame B revealed that the steepest gradients corresponded to signal increases from zero to the maximum value within 1 to 1.5 mm. If such a gradient falls within the measuring volume of the Raman measurements ($l = 0.6$ mm, diam $= 0.6$ mm), the estimated temperature differences within the probe volume are 600–1200 K, and spatial averaging effects become large. Furthermore, significantly smaller gradients occurred which were expected to stem from pure mixing of hot and cold gases. These gradients are not of importance with respect to spatial averaging. Evaluation of the OH PLIF images at $h = 10$ mm showed that the probability of finding a steep OH gradient within the Raman probe volume is approximately 7.5%. Thus, spatial averaging effects cannot be excluded and have to be considered in the discussion of the Raman results.

2.4. NO sampling probe measurements

The NO concentrations in the exhaust gas were measured by a sampling probe in the exhaust tube of the combustor. The measuring principle of the commercial analyzer (Ecophysics CLD700 EL HT) is based on a chemiluminescence measurement of NO_2^*, which is generated in the reaction of NO with O_3.

3. Results and discussion

3.1. Summary of global flame behavior

The flame structures and the mean values and fluctuations of the velocities, temperature, major species concentrations, and mixture fraction were discussed in the preceding article [1] and are summarized only briefly here. The flow fields of all three flames are characterized by a strong irz, which likely extends down into the central air nozzle, the injected flow of fresh gases which forms a cone with an opening angle of approximately 26°, and an outer recirculation zone (orz), as schematically displayed in Fig. 1. Flames B and C have almost identical mass flows and their mean velocity fields are very similar. The flow velocities of flame A are typically 3.5 times higher, but the overall shape of the flow field resembles those of the other flames. The flames are not attached to the fuel nozzle and are stabilized, on average, at $h \approx 5$ mm in flame A, $h \approx 4$ mm in flame B, and $h \approx 6$ mm in flame C. The flames are therefore partially premixed before ignition. The mean values further revealed that mixing is very fast and that burnout, as deduced from CH PLIF images, is completed at $h \approx 50$, 20, and 40 mm in flames A, B, and C, respectively. The highest mean temperatures (flame A: 1827 K, flame B: 1969 K, flame C: 1551 K) were measured within the

irz where the mean mixture fractions lie above the global values for f. Single-pulse OH and CH LIF images revealed that the instantaneous flame behavior is dominated by random turbulent fluctuations, at least in the near field of the flames [34]. Comparison of the measured temperatures and the adiabatic flame temperatures calculated for the instantaneous gas compositions further indicated that effects of turbulence–chemistry interactions play an important role over large regions of the flame. Finally, the results from the preceding article showed that flame B exhibits a different flame shape compared with flames A and C in that it is flatter and shorter. This behavior is closely related to the pronounced thermoacoustic oscillations of this flame. Phase-correlated measurements revealed that the irz and orz are subject to periodic variations of their size, which leads to an additional mixing process of fuel, air, and exhaust gas [25].

3.2. Correlation between temperature and mixture fraction

Fig. 2 illustrates the correlations between temperature and mixture fraction for the three flames at $h = 5$ mm, which was the lowest measuring height for the Raman measurements. Each symbol represents the result of a single-pulse measurement recorded at various radial locations from $r = 0$ to $r = 30$ mm. The solid lines represent adiabatic equilibrium and the result of a strained laminar counterflow diffusion flame calculation for the Tsuji flame geometry with strain rate $a = 400\ \mathrm{s}^{-1}$ [35,36]. The calculation was performed using the GRI 2.11 mechanism with unity Lewis number; the stoichiometric value of the scalar dissipation rate was $15.6\ \mathrm{s}^{-1}$. Although the partially premixed turbulent flames investigated cannot be directly compared with a counterflow diffusion flame, the calculated curves may serve here as an aid for assessing the thermochemical state of the flames under investigation. For all three flames, the scatter in mixture fraction reaches from $f = 0$ (pure air) to $f \approx 0.2$ (equivalence ratio $\Phi = 4.3$), with the majority of the samples between 0 and 0.1. This result reflects the fast mixing achieved with this nozzle configuration. In all three flames, there are a large number of samples with $T \approx 300$ K, even around $f_{\mathrm{stoich}} = 0.055$, which have not been ignited yet. This result suggests that the flames should be classified as partially premixed. The samples with the highest temperatures stem from the irz (red symbols) and are scattered around $f \approx 0.043$, $T \approx 2050$ K in flame A, $f \approx 0.052$, $T \approx 2100$ K in flame B, and $f \approx 0.043$, $T \approx 1850$ K in flame C. These samples lie quite close to the calculated curves; i.e., they are almost completely reacted. Their f values are larger than the global f values, reflecting that fuel is preferentially transported to the flame

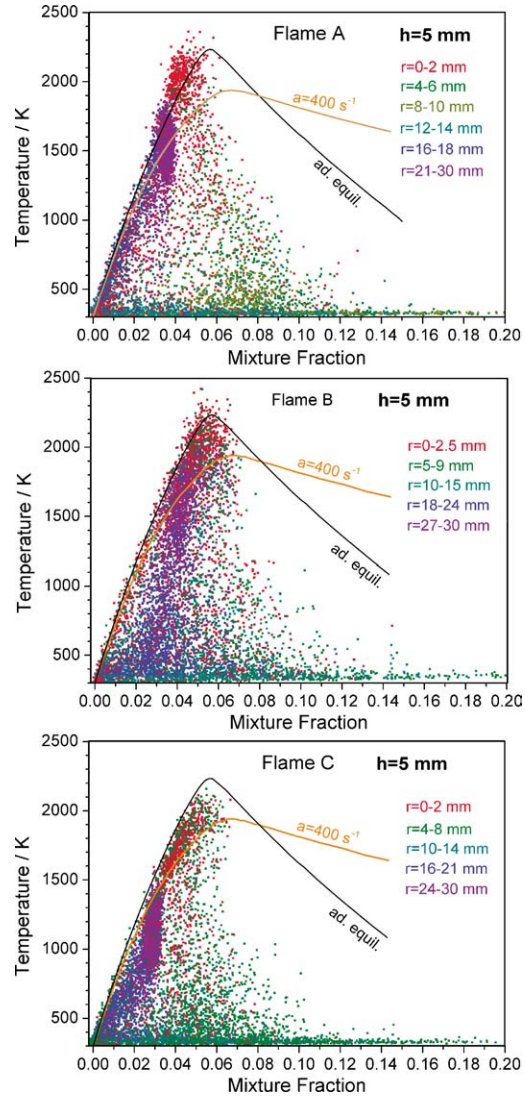

Fig. 2. Correlation between temperature and mixture fraction for the three flames at $h = 5$ mm. The different-colored symbols indicate different radial regions.

axis, thereby increasing the flame temperature above $T_{\mathrm{glob\ ad}}$ for the global mixture fractions (see Table 1). Considering that the irz generally plays an important role in the stabilization of swirl flames, the relatively high f and T values achieved by this nozzle configuration additionally increase the stabilizing effect of the irz. The samples from the orz (violet) exhibit significantly lower values of f and T and are scattered around $f \approx 0.035$, $T \approx 1250$–1800 K in flame A, $f \approx 0.042$, $T \approx 1200$–1850 K in flame B, and $f \approx 0.028$, $T \approx 800$–1400 K in flame C. Their f values show little variation and are close to f_{global}. Compared with adiabatic equilibrium, their temperatures are reduced by up to 600 K. One reason for

this is incomplete energy conversion: the results from the CH_4 measurements revealed that a few percent methane can be found in this overall fuel-lean region (see Fig. 7, discussed later). The second reason is that at $h = 5$ mm, the flame gases of the orz are quite close to the relatively cold burner plate and heat loss due to wall contact is expected. The temperature of the burner plate (made of stainless steel) was measured by a thermocouple at a position 4 mm below the surface of the plate and at $r \approx 30$ mm. Here, the temperatures of flames A, B, and C were 840, 650, and 600 K, respectively.

In addition to the single-shot probes in Fig. 2 discussed so far, there is a large number of samples with intermediate temperatures mainly from the radial region $r \approx 2$–12 mm, i.e., from the shear layer between the irz and the main stream of the injected gases. These partially reacted mixtures can be explained by two different effects of turbulence–chemistry interaction: The first is local flame extinction. There are several indications that localized flame extinction occurs. The single-shot 2-D LIF images of CH that were used to visualize the reaction zones revealed that the CH layers were typically connected over several centimeters, but sections with very little or no CH were also found, which could indicate events of local flame extinction [34]. Furthermore, the flames were very noisy and burned unsteadily. Also, it is generally known from measurements in non-premixed flames that local flame extinction events lead to gas mixtures with intermediate temperatures that are far from adiabatic equilibrium [2,4,20,37]. The second effect is ignition delay. The fast mixing of fuel, air, and exhaust gas in flames with recirculation can lead to a large variety of gas compositions and temperatures. The reaction of such mixtures is subject to ignition delay with a time constant that depends on the local temperature and composition. For gas mixtures and temperatures typical of the flames investigated here, the ignition delay time was calculated using the SENKIN code (CHEMKIN II) with the GRI 3.0 mechanism. The mechanism was extended as described by Schelb et al. [38,39] by adding a submodel that contains reactions of mainly CH_3O_2 and CH_3NO_2, which were found to improve the performance of the mechanism at temperatures below approximately 1200 K. To have well-defined initial conditions for the calculation, the gas compositions and temperatures were determined as follows: For the three mixture fractions $f = 0.031$, 0.042, and 0.055 (corresponding to $\Phi = 0.55$, 0.75, and 1.0), the adiabatic equilibrium composition, including NO and the radicals O, H, and OH, and temperature were calculated. In the next step, several compositions of burnt gas (at adiabatic equilibrium) and fresh gas (at 300 K, same f) were calculated in such a way that the resulting temperatures of the mixtures were 1000, 1250, 1500, and 1750 K. These compositions were taken as the input data for the SENKIN calculation. The results are displayed in Fig. 3. As can be seen, the ignition delay time depends strongly on T, but changes only little with the mixture fraction in the f range 0.031 to 0.055. To compare the delay times with the time scale τ of the flow field, τ can be roughly estimated by $\tau \approx h/u$, where h is taken as the measuring height ($\approx$5 mm) and u as a typical flow velocity of $\approx$10 m/s. This leads to $\tau \approx 0.5$ ms, with, of course, a large scatter due to the turbulent nature of the flame. Thus, ignition delay can significantly contribute to the occurrence and residence time of partially reacted mixtures, especially at temperatures below 1500 K. In the authors' opinion, both effects—local flame extinction and ignition delay—contribute to the observed samples with intermediate temperatures. It should be noted that a similar result was obtained in CH_4/air jet flames with swirling coflow [20].

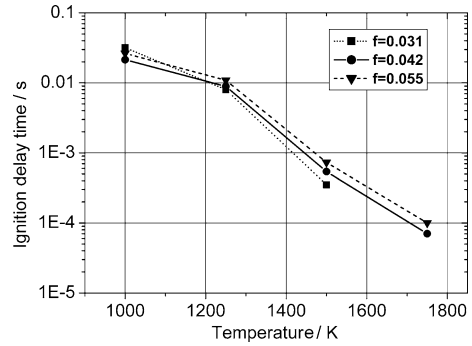

Fig. 3. Ignition delay times for mixtures of cold CH_4–air and adiabatic exhaust gas.

In this context, a possible contribution from spatial averaging effects of the measurement should be addressed. As stated in Section 2.3, the probability of finding a steep OH gradient within the Raman probe volume at $h = 10$ mm is approximately 7.5%. In comparison, the fraction of samples with partially reacted mixtures at $h = 10$ mm, as deduced from the single-shot Raman measurements, is on the order of 35%. Thus, spatial averaging effects may contribute to up to about 20% of the observed partially reacted mixtures. However, the majority of the samples with intermediate temperatures has to be attributed to real effects of turbulence–chemistry interaction as described above. In a further test, Raman measurements with a reduced measuring volume of $0.3 \times 0.3 \times 0.6$ mm were performed (by decreasing the width and height of the slit of the spectrograph). The resulting PDFs of the mixture fraction showed no significant differences from those obtained with the usual size of the probe volume. The PDFs of temperature yielded a similar result, but they exhibited a larger scatter at high temperatures ($T \approx 2000$ K), which is explained by the

lower precision of the measurements with only ≈25% of the usual probe volume. Thus, this test also gave no indication that spatial averaging effects would significantly alter the interpretation of the scatterplots.

Overall, the scatterplots from the three flames have a quite similar appearance, and the observed differences are due mainly to the different global air/fuel ratio and mixing process. Because of the higher exit velocity of flame A, the time for mixing before reaching $h = 5$ mm is here significantly shorter than for the other two flames. This explains the slightly richer mixtures in the radial region near the fuel injection ($r \approx 5$–10 mm). Due to its high air/fuel ratio, flame C exhibits significantly fewer samples with stoichiometric or fuel-rich mixtures than flames A and B. A closer look at the scatterplots reveals that the samples from the different radial regions displayed by different colors are not as well separated in flame B as in the other two flames. This is caused by the thermoacoustic oscillations and the periodic variation of the size of the recirculation zones, which leads to an overlap of different flow field regions during measurements without phase locking. Also, the periodic variations induce an additional mixing process by the pumping-like movement of the recirculation zones [1,25].

Preheating of the injected air by heat exchange at the nozzle surface is also worth mentioning. After a short time of operation, the exhaust pipe of the combustion chamber becomes red hot and the windows reach a temperatures up to 1000 K. The nozzle does not come into contact with hot combustion gas but might be subject to heat transfer via radiation. For a correct characterization of the boundary conditions, the preheating of the injected air was determined. The temperature of the air can be deduced from T–f scatterplots at $h = 5$ mm when looking at $f = 0$. However, more sensitive and accurate is the correlation between T and H_2O mole fraction (see Fig. 9, discussed later). Because dry air was supplied to the burner, pure air from the nozzle is water-free. Inspection of temperatures from samples with $X(H_2O) = 0$ revealed that for flame A, the air temperature scatters between 290 and 360 K; for flame B, between 290 and 380 K; and for flame C, between 305 and 355 K. There is, thus, a small degree of preheating with some scatter due to variations of the heat transfer between a fluid element and the surface of the nozzle.

Clearly, mixing and reaction proceed with increasing height, as can be seen in the scatterplots of temperature at $h = 15$ mm (Fig. 4). Here, significant differences between the three flames are apparent. In flame B, the majority of the samples exhibit temperatures close to the calculated curves and the burnout is almost complete. In contrast, the scatterplots of flames A and C are still dominated by partially reacted mixtures. While in flame A a lot of these samples are found around $f_{\text{stoich}} = 0.055$, they are significantly shifted to smaller f values ($f \approx 0.028$, $\Phi \approx 0.5$) in flame C. Here, most of these samples are located in the outer region of the flame ($r > 20$ mm), in the orz and the neighboring shear layer between the orz and the airflow from the outer air nozzle. The reaction progress of these gases is relatively slow because the mixtures are very lean and mixing from hot gas from the irz, which would promote the reactivity, is hindered by the gap between the irz and the orz.

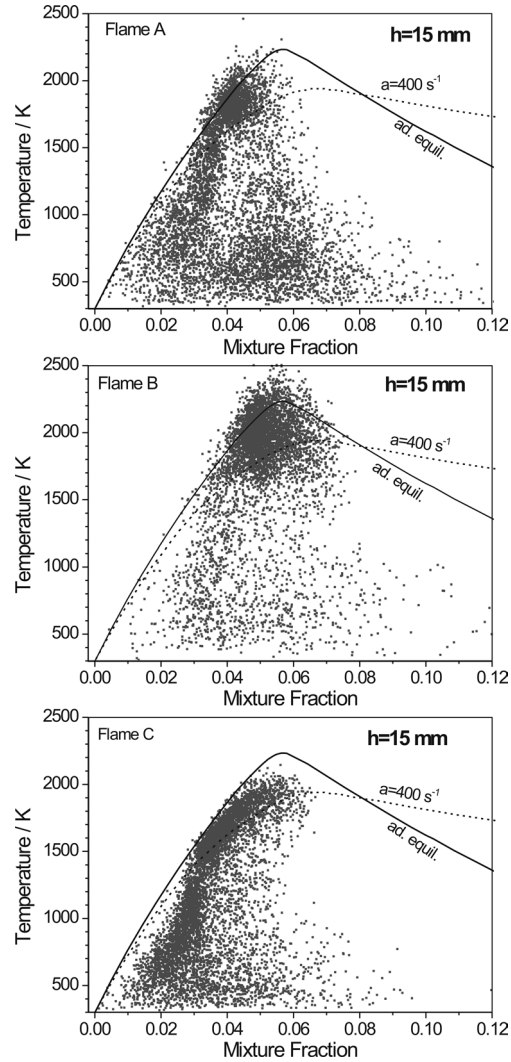

Fig. 4. Correlation between temperature and mixture fraction at $h = 15$ mm. In contrast to Fig. 2, the scale for the mixture fraction axis is reduced to $f = 0.12$.

The scatterplots at $h = 30$ mm are displayed in Fig. 5. Flame B has reached a homogeneous temperature and mixture fraction distribution for all radial locations, and the thermochemical state is close to adiabatic equilibrium. In flame A, the scatter in f

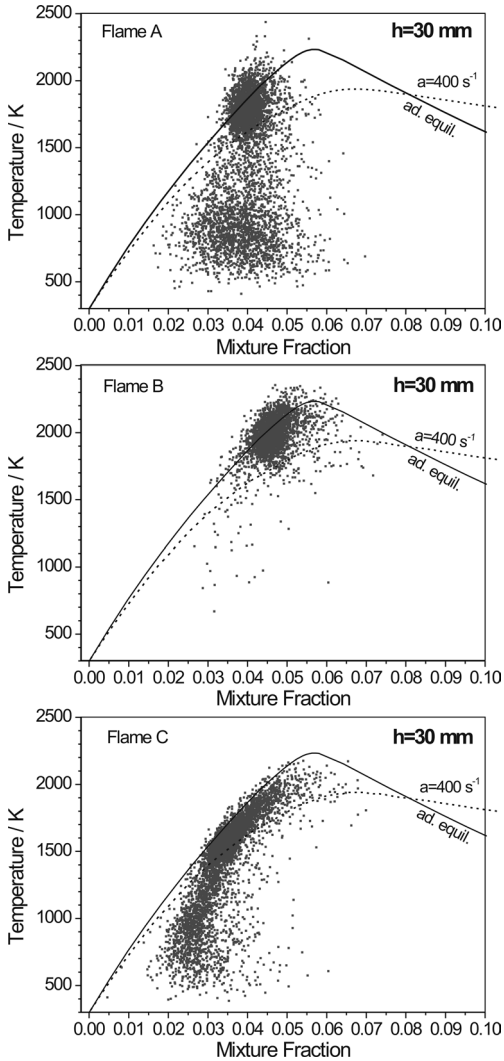

Fig. 5. Correlation between temperature and mixture fraction at $h = 30$ mm. In contrast to Fig. 2, the scale for the mixture fraction axis is reduced to $f = 0.10$.

has become small but the temperature distribution still shows the separation into completely reacted samples near the axis and partially reacted mixtures further outside. In flame C, mixing is slower in comparison to the other two flames, and there remains a large number of samples with intermediate temperatures in the orz. Finally, at $h = 60$ mm (not shown), flames A and C have also reached a thermochemical state close to equilibrium. The rapid reaction progress in flame B was addressed in the preceding article [1] and attributed to the thermoacoustic oscillations. Periodic movement of the recirculation zones generates a more intense mixing of fresh gas and exhaust gas, and the admixture of hot combustion gases to unburned gases leads to higher reactivity and flame propagation [25].

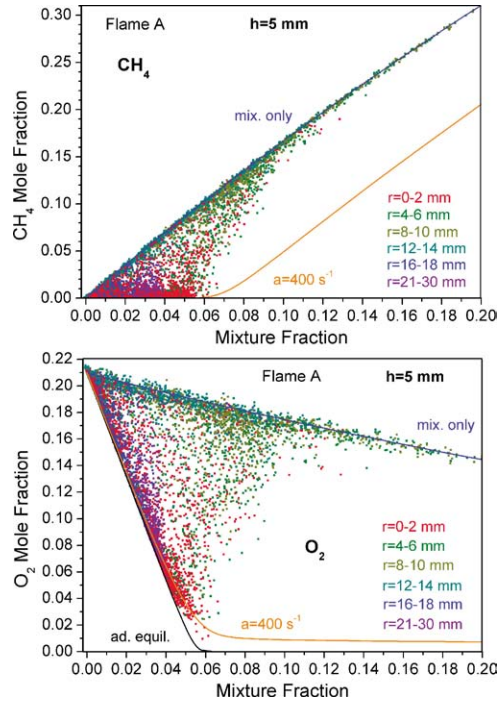

Fig. 6. Correlation between CH_4 and mixture fraction and of O_2 and mixture fraction at $h = 5$ mm in flame A.

3.3. Species mole fractions

The scatterplots of species mole fractions are, to a large extent, in agreement with the scatterplots of temperature and the above-given explanations. Sample results of CH_4 and O_2 scatterplots are shown in Fig. 6 for $h = 5$ mm in flame A. Here, a calculated curve for pure mixing of CH_4 and air is also displayed as a limiting case. As in the scatterplots of temperature, it can be seen that the samples comprise non-reacted, partially reacted, and completely reacted gas mixtures. A few samples reach CH_4 mole fractions up to $X(CH_4) = 0.4$ but more than 99% are below $X = 0.3$. The fuel-rich samples do not follow the calculated curve for strained laminar flames and are close to the curve "mixing only." The same is valid for the other two flames, in accordance with the $T-f$ scatterplots. The scatterplots of CH_4 and O_2 also provide evidence that most of the samples from the radial region $r \approx 4$–14 mm follow the curve "mixing only," demonstrating the high degree of premixing. The correlation between CH_4 and O_2 mole fractions, shown in Fig. 7 for flame A at $h = 5$ mm, directly shows the coexistence of fuel and air. Here, samples with $X(CH_4) \approx 0$ are completely reacted, and all others follow the curve of pure mixing or are partially reacted.

The mole fractions of the intermediate species CO and H_2 reach the highest values in flame B, as is ex-

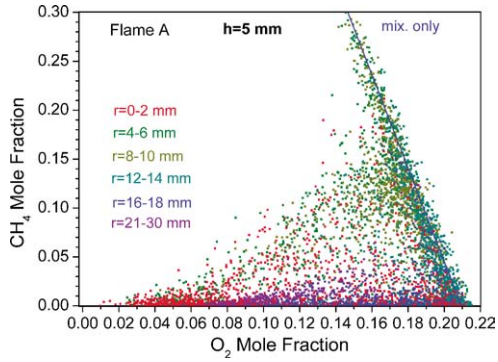

Fig. 7. Correlation between CH_4 and O_2 at $h = 5$ mm in flame A.

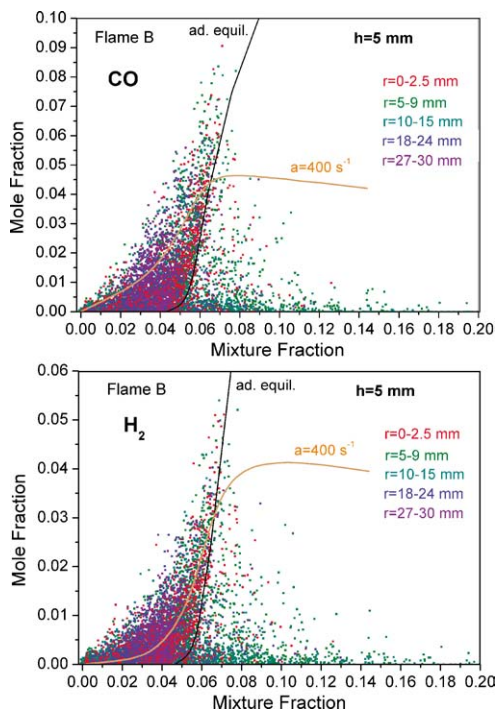

Fig. 8. Correlation between CO and mixture fraction and of H_2 and mixture fraction at $h = 5$ mm in flame B.

pected from the air/fuel ratio. Fig. 8 illustrates the CO and H_2 scatterplots in this flame at $h = 5$ mm. For these relatively low concentrations, the single-shot measuring uncertainty is on the order of 10–30% and responsible for some of the observed scatter. It is seen that CO and H_2 tend to follow the calculated curves for $f < 0.06$ but not on the fuel-rich side. However, as observed before, there are no completely reacted mixtures on the fuel-rich side, and the significant CO and H_2 mole fractions observed in the few fuel-rich samples are due to mixing.

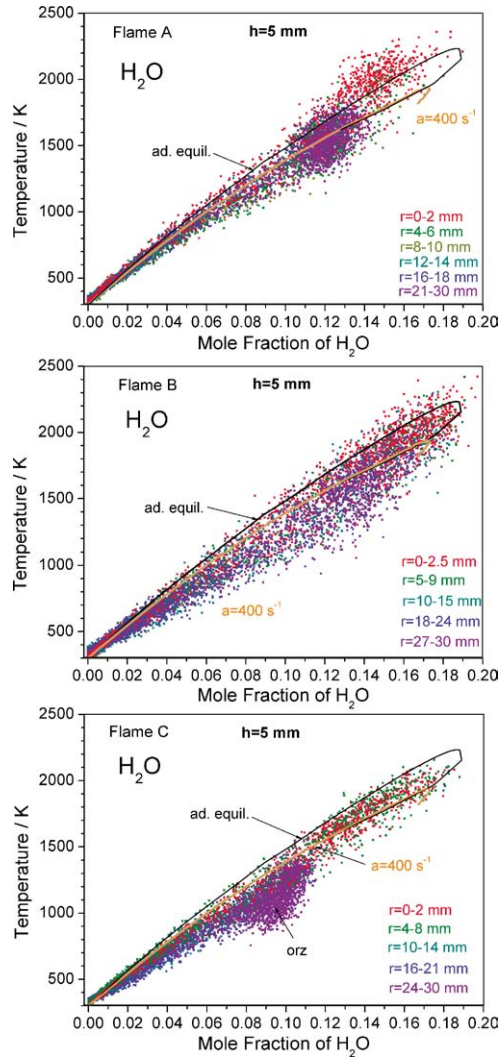

Fig. 9. Correlations between temperature and H_2O mole fraction for the three flames at $h = 5$ mm.

Correlations between temperature and H_2O mole fractions are illustrated in Fig. 9 for the three flames at $h = 5$ mm. Here, the calculated curves are drawn only for $f = 0$–0.1, i.e., for fuel-lean and near-stoichiometric mixtures, because there are only few samples with $f > 0.1$ (which are at room temperature). It can be seen that the measured results are in reasonable agreement with the strained laminar flame calculation with $a = 400\ \mathrm{s}^{-1}$. The samples from the irz are better matched by the adiabatic equilibrium curve, which is explained by the relatively long residence time of the recirculating gas at high temperatures. For flame A, the agreement between the measured results and the curve with $a = 400\ \mathrm{s}^{-1}$ is good. For flames B and C, the measured temperatures lie below the calculated temperatures, especially for

the samples from the orz and the shear layer between the orz and the neighboring airflow. As mentioned before, these samples have probably experienced some heat loss by contact with the burner plate so that their temperature is reduced. In flame A, the heat loss is expected to be less pronounced because of the higher thermal power and the shorter residence time of the gas in the combustion chamber. As in the $T-f$ scatterplots of Fig. 2, it is seen that the samples from different radial regions, e.g., from the orz (violet), scatter much more strongly in flame B in comparison to the other two flames, indicating that the different regions within the flame are not as well separated in this flame. Overall, the correlations between temperature and H_2O mole fraction exhibit much less scatter than the $T-f$ correlation. The close correlation between T and H_2O mole fraction reflects that both quantities track reaction progress, but heat loss effects cannot be neglected in the outer region of the flames.

The correlations between T and CO_2 mole fractions at $h = 5$ mm are displayed in Fig. 10. Again, the calculated curves are restricted to $f \leqslant 0.1$. It is seen that samples from the irz (red) and the adjacent shear layer (green) are in reasonable agreement with adiabatic equilibrium and that the calculation with $a = 400\ \mathrm{s}^{-1}$ yields much poorer agreement, especially for high CO_2 mole fractions. Samples from the orz (violet) and the shear layer between the orz and inlet flow of air (blue) exhibit significant deviations from the calculations in the way that the measured CO_2 mole fractions are larger than those in laminar flames or that the measured temperatures are lower. The discrepancies may, at least partly, be explained by a temperature drop in these samples. The authors are, however, not sure whether effects based on chemical kinetics might also contribute to the observed discrepancies. It should be noted that the correlations between H_2O and CO_2 (not shown) are in quite good agreement with adiabatic equilibrium and strained laminar flame calculations with low or medium strain rates.

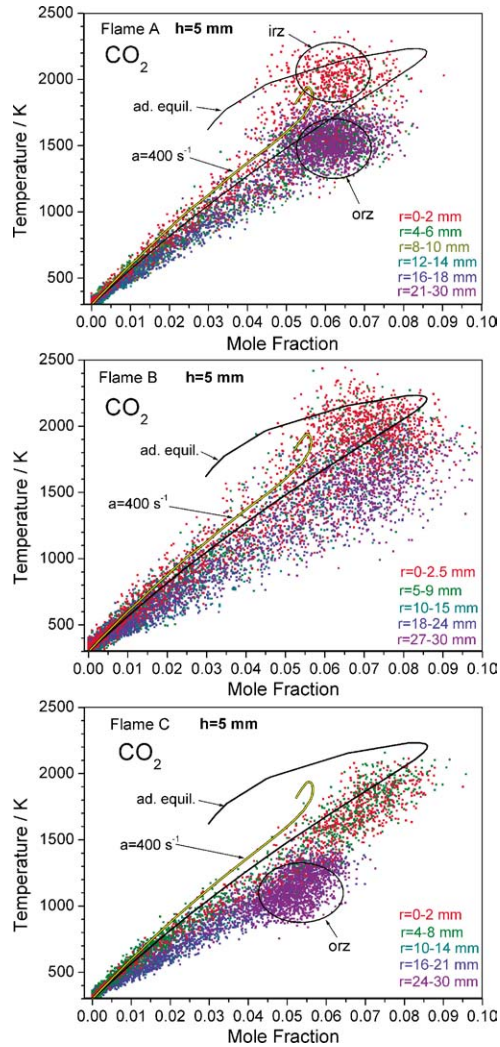

Fig. 10. Correlations between temperature and CO_2 mole fraction for the three flames at $h = 5$ mm.

3.4. Flame stabilization and mixing

The averaged CH LIF distributions shown in the preceding article [1] demonstrated that ignition and flame stabilization occur at $h \approx 5$ mm for flame A, $h \approx 4$ mm for flame B, and $h \approx 6$ mm for flame C. The scatterplots at $h = 5$ mm (Fig. 2) revealed some differences in mixing and gas composition for the three flames in this region which may influence the ignition. To demonstrate the differences more clearly, the corresponding histograms of f for the three flames are shown in Fig. 11. Here, only samples with low temperatures ($T < 1000$ K), i.e., gas mixtures that have not been ignited or completely reacted yet, are considered. The criterion for conditioning, $T = 1000$ K, is somewhat arbitrary; however, a slightly different limit would yield a similar result. In interpreting the histograms, one must consider that flame stabilization takes place preferentially under near-stoichiometric conditions where the flame speed and reactivity reach a maximum. Comparison of the three histograms shows that flame C exhibits fewer near-stoichiometric and rich mixtures at this height than the other two flames. This is not surprising because flame C is operated at the lowest fuel/air ratio. However, considering the stabilization height of flame C is above $h = 5$ mm, this means that the majority of the mixtures have to be ignited under lean, less favorable conditions. The situation is different for flames A and B: flame A exhibits a broad distribution

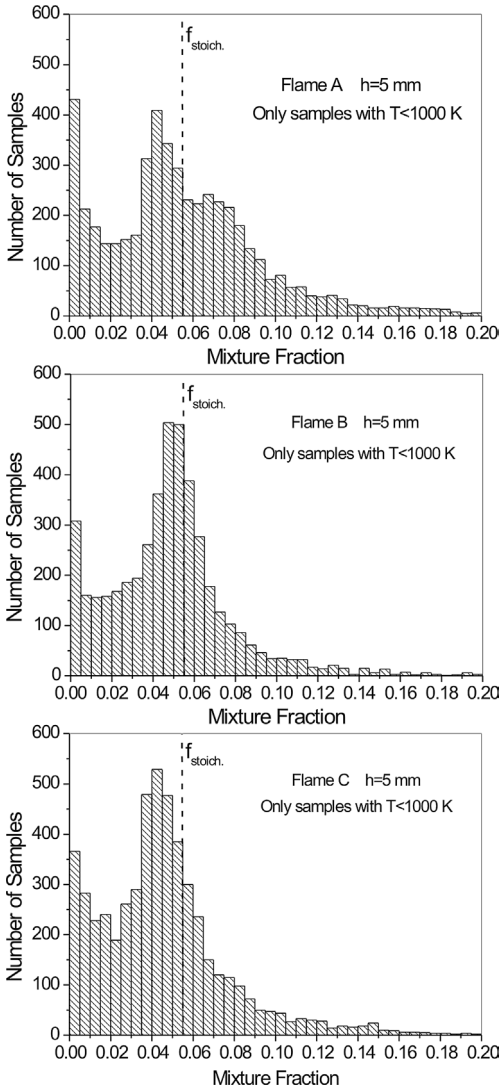

Fig. 11. Histograms of mixture fraction for the three flames at $h = 5$ mm. Only samples with $T < 1000$ K are included.

around f_{stoich} and flame B has its maximum close to f_{stoich}. Besides the generally lower temperature level of flame C, the rapid mixing producing lean mixtures before ignition is another reason why flame C is more unstable than the other two flames.

3.5. NO concentrations in the exhaust gas

The NO concentrations measured by a sampling probe in the exhaust gas tube were 6, 25, and 9 ppm for flames A, B, and C, respectively. The relatively high concentrations in flame B might be explained by the high temperature level (see Table 1). The different NO levels in flames A and C cannot be explained by the temperature, because the temperature level of flame A lies roughly 180 K above that of flame C. However, due to the different flow velocities, the residence time in flame A is much shorter than that in flame C, which should decrease the formation of thermal NO. A discussion on the influence of unsteady combustion behavior on NO formation is beyond the scope of this article, and the NO concentration measurement is presented mainly as another quantity for comparison with results from flame calculations.

4. Summary and conclusions

Single-shot laser Raman measurements were performed in three swirling CH_4/air flames operated in a gas turbine model combustor at atmospheric pressure. While results about flame structures, flow velocities, and averaged distributions of temperature, species concentrations, and mixture fraction have been reported previously [1], the current article focuses on effects of mixing and turbulence–chemistry interactions.

The flames were stabilized approximately 10 mm above the fuel nozzle, which corresponds to $h \approx$ 5 mm above the burner plate. Here, CH_4 and air have reached a high degree of premixing ($0 \leqslant f \leqslant 0.2$) so that the flames should be classified as partially premixed. Within the inner recirculation zone, the gas mixtures were mostly completely reacted and relatively fuel-rich ($f > f_{\text{glob}}$), and their thermochemical states were close to adiabatic equilibrium. In the outer recirculation zone, the temperatures were lower than in the irz due to leaner mixtures and to the presence of small amounts of CH_4 that had not yet reacted in this overall lean environment. In addition, the results indicated a temperature reduction of the gas mixtures in the outer region of the flame due to heat conduction to the burner plate. In the radial regions where the fresh gas was injected and in the neighboring shear layer built with the irz, most of the samples were unreacted or partially reacted. The most probable explanation for the occurrence of partially reacted mixtures was local flame extinction or ignition delay. These mixtures were characterized by intermediate temperatures and the coexistence of CH_4 and O_2. The correlations of the H_2O and CO_2 mole fractions and temperature were in reasonable agreement with a strained laminar flame calculation. The scatterplots further showed that mixing and reactions were completed at $h \approx 50$ mm for flame A, $h \approx 20$ mm for flame B, and $h \approx 45$ mm for flame C.

Flame C was operated near the lean extinction limit and burned unstably with random fluctuations of the flame stabilization position. The obvious reasons for the unstable ignition lay in the relatively low temperature of this flame ($T_{\text{glob ad}} = 1570$ K) and in the

fact that most of the mixtures were already fuel-lean before they reach the stabilization region. The flame could probably be operated at an even lower fuel/air ratio if the nozzle configuration and the flow field were modified in such a way that the hot gases from the irz reached the region with near-stoichiometric mixtures. However, this was not pursued in our studies. Flame B had a mean velocity field similar to that of flame C, but exhibited a quite different flame shape. The flame zone was rather flat, and burnout was achieved at a significantly lower height than in the other flames. This behavior is related to the thermoacoustic oscillations of flame B which were found to influence the mixing process.

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